

CNN-Based Mammographic Breast Cancer Detection with Enhanced Accuracy Using Pectoral Muscle Removal

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Abstract:

Breast cancer continues to be a major global health concern, making early and reliable diagnosis essential for improving patient outcomes. Mammography is the most widely used screening method, yet the presence of the pectoral muscle, particularly in mediolateral oblique (MLO) views, often interferes with accurate analysis. The muscle region exhibits texture patterns that resemble tumours, which can mislead automated systems and reduce classification accuracy. This work presents an enhanced breast cancer detection approach that integrates pectoral muscle removal with a Convolutional Neural Network (CNN) classifier. The proposed preprocessing method applies edge detection and Hough-based line estimation to accurately identify and suppress the pectoral muscle while preserving relevant breast tissue. Experimental evaluation demonstrates that eliminating the pectoral muscle significantly improves the model's sensitivity and overall classification accuracy, even in low-quality or blurred mammograms. The results highlight the importance of effective preprocessing in strengthening deep learning performance and confirm the potential of the proposed pipeline for reliable computer-aided breast cancer diagnosis.

Key Words: Mammogram, Cancer Detection, CNN, Hough lines, Pectoral Muscle, Edge Detection.

1. INTRODUCTION

Breast cancer continues to be one of the leading health concerns worldwide, with cases rising every year. In women, especially, the disease commonly appears due to abnormal and uncontrolled growth of breast tissue cells. Mammography remains the most widely used clinical method for early detection because it is inexpensive and non-invasive. However, accurate interpretation usually requires trained radiologists, and manual analysis is prone to oversight. Deep learning models, particularly Convolutional Neural Networks (CNNs), have recently demonstrated remarkable potential in automating cancer detection. Yet, one major challenge is the presence of the pectoral muscle in mammograms, mainly in mediolateral oblique (MLO) views. Since the muscle has texture patterns similar to cancerous tissues, CNNs may misclassify these regions and reduce diagnostic accuracy. The aim is to determine whether low-quality or blurred mammograms can still be used for cancer detection, provided the pectoral muscle is accurately removed.

2. LITERATURE REVIEW

Heath et al. (2000) developed and described the Digital Database for Screening Mammography (DDSM), which provided thousands of labelled mammograms with verified benign and malignant cases. This dataset laid the foundation for many later CAD systems and remains a benchmark for evaluating mammogram classification algorithms.

Krizhevsky et al. (2012) successfully demonstrated the power of Convolutional Neural Networks (CNNs) on large-scale image classification tasks. Although their work was not specific to medical imaging, it inspired subsequent studies to apply CNNs to mammograms and other medical modalities due to their ability to automatically learn hierarchical features.

Zhou et al. (2016) introduced Class Activation Mapping (CAM), and later Grad-CAM was proposed as a gradient-based extension to visualize important regions in an image for any CNN-based model. These methods allow practitioners to see which parts of the image most influenced a prediction. In the context of mammography, Grad-CAM becomes a powerful tool for verifying whether the model is focusing on tumor regions or irrelevant areas such as the pectoral muscle.

3. OBJECTIVES / AIMS

The objective of this project is to design and implement an intelligent computer-aided system for the accurate detection of breast cancer using mammogram images. A key objective is to enhance the quality of mammograms through effective preprocessing techniques, enabling clearer visualization of diagnostically significant regions. Special emphasis is placed on identifying and eliminating the

pectoral muscle present in mediolateral oblique mammograms, as this region often exhibits intensity patterns similar to malignant tissue and can mislead automated classifiers.

4. RESEARCH METHOD / METHODOLOGY

Grayscale Conversion

The mammogram is converted to grayscale for simplifying intensity-based edge detection.

Edge Detection (Canny Operator)

Performs smoothing, gradient calculation, non-maximum suppression, double thresholding, and edge linking.

Hough Line Detection

Uses parametric space to detect straight boundary lines of the pectoral muscle.

Angle range used: 10° – 70° , Distance range: 5–200 pixels

Muscle Removal

Once the boundary line is detected, the region is removed/masked to create a clean breast region.

5. FINDINGS / RESULTS

The experimental analysis of the proposed mammographic cancer detection system produced several significant findings that highlight the importance of preprocessing, particularly pectoral muscle removal, in improving deep learning based diagnosis. The muscle region often appears bright and structurally similar to malignant tissue. After applying the muscle-removal technique using Canny edge detection and Hough line transformation, the CNN's classification stability improved noticeably. The segmentation process successfully isolated the muscle boundary in the majority of MLO images, even when varying levels of blur were introduced and indicates that eliminating irrelevant anatomical structures allows the CNN to focus solely on actual breast tissue, thereby enhancing feature extraction and decision-making. Sensitivity and specificity values also improved after preprocessing, showing that both malignant detection and benign differentiation became more consistent.

6. DISCUSSION / ANALYSIS

The results obtained from this study highlight the strong influence of preprocessing on the effectiveness of deep learning models used for mammographic breast cancer detection. One of the most important observations is that the presence of the pectoral muscle significantly affects CNN

training and prediction outcomes. Because the muscle region appears with high intensity and structural similarity to malignant tissue, models trained on raw mammograms tend to produce higher false-positive rates. This confirms that deep learning systems, while powerful, are sensitive to misleading anatomical features when preprocessing is inadequate.

7. RECOMMENDATIONS

- Pectoral muscle removal should be applied to all MLO-view mammograms before classification.
- Contrast enhancement techniques like CLAHE should be used to improve detection of subtle lesions.
- CNN models should be trained using diverse and well-annotated mammogram datasets.
- Explainable AI methods such as Grad-CAM are recommended to improve interpretability and clinical trust.

8. LIMITATIONS

- Automatic stage estimation is image-based and does not replace clinical or pathological staging.
- The system currently supports only binary classification (benign or malignant).
- Results may vary when tested on mammograms from different imaging devices.
- Clinical validation with radiologists is required before real-world deployment.

9. CONCLUSION / SUMMARY

This work demonstrates the effectiveness of a deep learning–based approach for automated breast cancer detection using mammogram images. By integrating structured image preprocessing with a Convolutional Neural Network (CNN), the system successfully addresses challenges commonly associated with mammographic analysis. In particular, the removal of the pectoral muscle significantly reduces false positives by eliminating misleading anatomical regions that closely resemble malignant tissue. Contrast enhancement further strengthens the visibility of subtle abnormalities, allowing the CNN to extract meaningful and discriminative features. The experimental results confirm that the proposed methodology improves classification accuracy and model reliability compared to unprocessed mammogram analysis. The inclusion of visual explanation techniques enhances interpretability and supports clinical confidence in automated predictions. Overall, the system offers a reliable and efficient solution for supporting early breast cancer detection and

demonstrates strong potential as a computer-aided diagnostic tool, especially in environments where expert radiological analysis may be limited

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